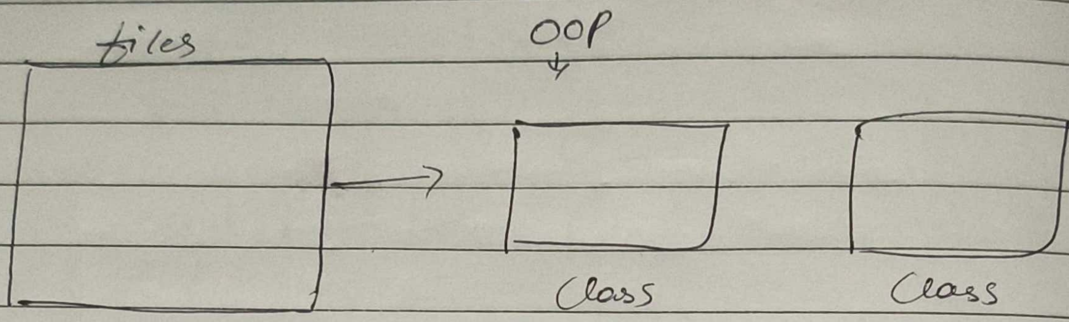
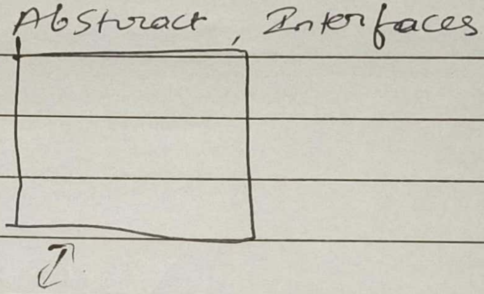
8/8/25
Day 28

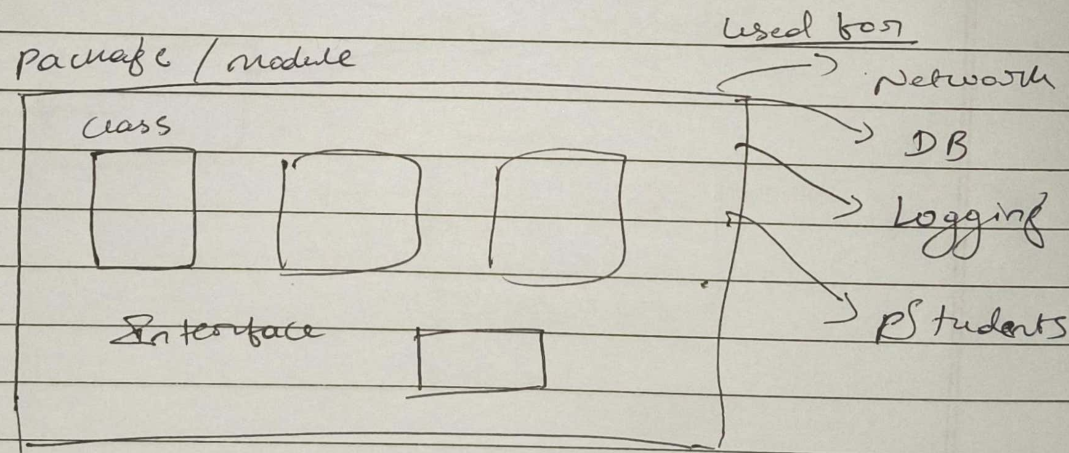
(3) Introduction To working with packages & modules

packages/modules :

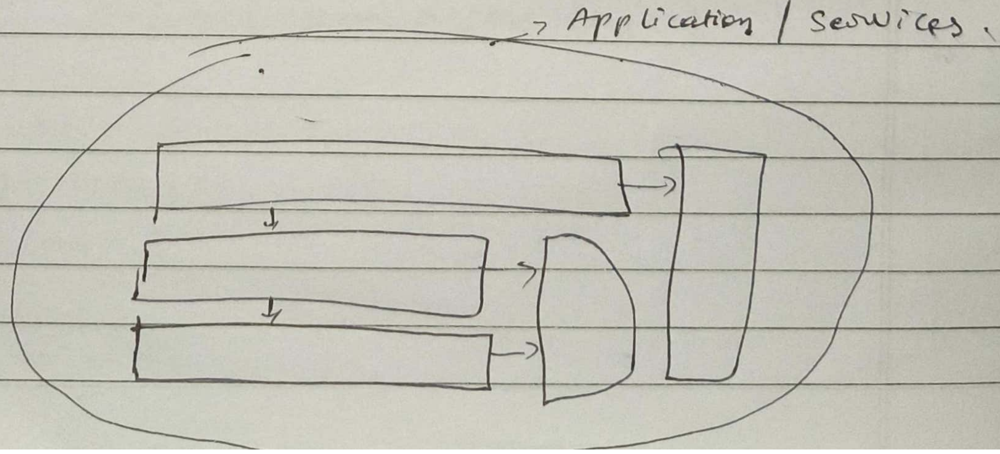


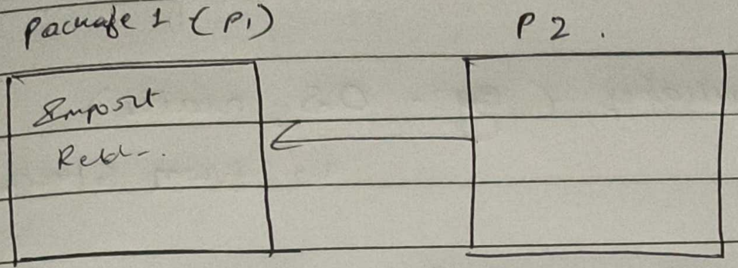


make folder & drop all in that → called as package/module



our Software :- block diagram.





13 Understanding pacuafes in python

1) Advantages of using pacuafes in python.

-> Modularity
-> Reusability
-> Name Space Manafement
-> Organization
-> Scalability.

2) Introduction to pacuages in python.

A pacuafe in python is a directory that contains multiple related modules (python files).

-> A pacuafes typically includes an

__init__.py file.

-> which allows the directory to be recognized as a pacuafe.

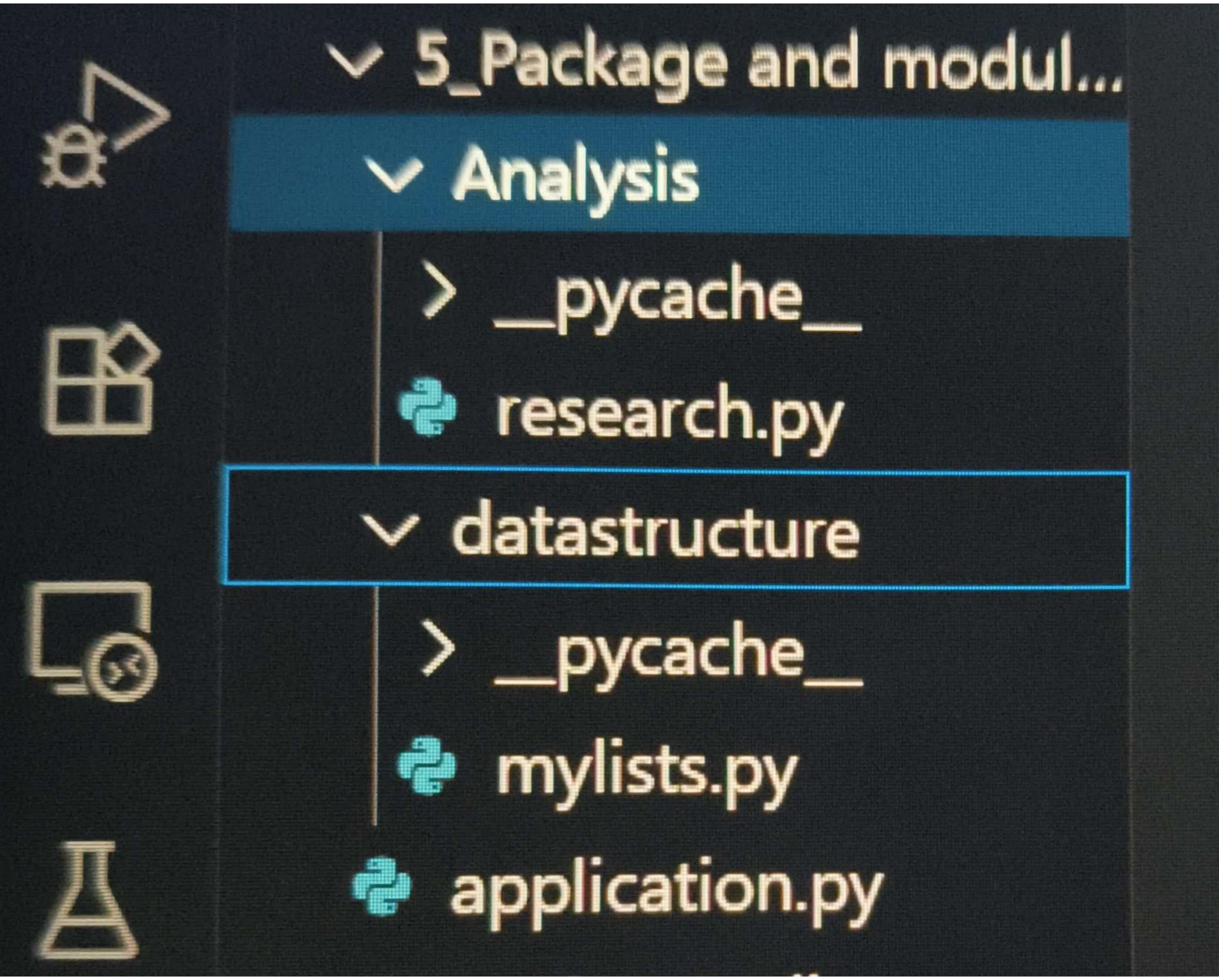
There are two main types of packages.

1) Built-in packages (eg:- OS, math)
↳ from standard library

2) User-defined packages that are created by developers to organize their own modules.

* Python folder structure.

```
my_project /
  ├── app.py
  └── datastructures /
        ↳ __init__.py
        ↳ mylist.py
```


5_Package and modul...
Analysis
__pycache__
research.py
datastructure
__pycache__
mylists.py
application.py

PersonClass.py ● VehicleClass.py application.py ×

5_Package and modules > Modules-Project > application.py > ...

```python
from datastructure.mylists import StudentList
from datastructure.mylists import MyList
from Analysis.research import MyAnalysis
print("hello")

new_list = StudentList()
new_list.add(1)
new_list.add(2)
new_list.add(3)
new_list.print()

new_list = MyList()
new_list.add(1)
new_list.add(2)
new_list.add(3)
new_list.print()

new_analysis = MyAnalysis()
new_analysis.add(1)
new_analysis.add(2)
new_analysis.add(3)
new_analysis.print()
```

Ln 17, Col 1 Spaces: 4

5_Package and modules > Modules-Project > datastructure > mylists.py > MyList

```python
class MyList:

    def __init__(self):
        self.list = []

    def add(self, item):
        self.list.append(item)

    def remove(self, item):
        self.list.remove(item)

    def print(self):
        print(self.list)

class StudentList:

    def __init__(self):
        self.list = []

    def add(self, item):
        self.list.append(item)

    def print(self):
        print(self.list)
```

5_Package and modules > Modules-Project > Analysis > research.py >

```python
class MyAnalysis:

    def __init__(self):
        self.list = []

    def add(self, item):
        self.list.append(item)

    def remove(self, item):
        self.list.remove(item)

    def print(self):
        print(self.list)
```